

Strawberry Production Cost Comparison

Prepared for Japfa Indonesia follow-up discussion | MANNA CEA | 12 July 2026

1. Purpose

This memo provides an indicative comparison of strawberry production cost in Indonesia versus Korea, using an indoor vertical strawberry farm model as the reference. The previous feasibility report was based on a 4,000 m² controlled-environment greenhouse concept; this memo adds a practical view for a compact indoor vertical production model.

2. Production Assumption

For indoor vertical strawberry production, the key productivity driver is: planting density x yield per plant = production per floor area.

Item	Indicative assumption
Facility type	Indoor vertical strawberry farm
Area basis	Floor area
Rack system	3-tier production system
Planting density	Approx. 19-20 plants/m ² of floor area
Conservative yield per plant	Approx. 0.6 kg/plant over extended indoor production
Resulting productivity	Approx. 15-16 kg/m ² /year

This production target is technically reasonable for a pilot and scale-up model, but it should be validated under local Indonesian operating conditions.

3. Indicative Cost Comparison

Cost item	Indonesia	Korea	Comment
Labor	Lower	Higher	Indonesia has a structural advantage in farm labor and local installation labor.
Electricity tariff	Generally lower	Higher	Indonesia may have lower tariff, but higher cooling and dehumidification demand.
Cooling / HVAC load	Higher	Moderate	Indonesia tropical climate requires continuous cooling and humidity control.
Seedlings / variety license	Higher risk	Lower risk	Indonesia may need import pathway, nursery development, and licensing structure.
Local construction	Lower to moderate	Higher	Local sourcing can reduce cost if quality is controlled.
Technical operation risk	Higher early stage	Lower	Korea has mature strawberry growers, suppliers, and maintenance ecosystem.

4. Indicative Production Cost Range

The following range is a preliminary business-planning estimate, not a final quotation:

Country	Stabilized production cost estimate
Indonesia	USD 8-12/kg
Korea	USD 11-16/kg

Indonesia is expected to be more cost-competitive after operational stabilization, mainly because of lower labor and local installation costs. However, this advantage can be reduced by higher cooling load, imported equipment

dependency, seedling logistics, and early-stage technical training cost.

5. Practical Conclusion

- Indonesia can potentially produce premium Korean-variety strawberries at a lower cost than Korea, but only after the farm reaches stable operation.
- The first pilot should validate actual yield per plant, electricity use, reject / waste / unsold ratio, and achievable offtake price by grade and channel.
- For a bankable financial model, MANNA CEA recommends applying a 10-15% loss allowance during the first operating cycle and reducing this to 5-8% after stabilization, subject to actual performance and offtake conditions.

6. Data Required for Final Model

- Confirmed facility format and floor area
- Number of tiers, bed length, and planting density
- Local electricity tariff and demand charge, if any
- Labor plan and salary level
- Local HVAC design and cooling-load estimate
- Seedling source, variety license, and nursery plan
- Target sales channel and offtake price by grade
- Packaging, cold-chain, and distribution cost

Note: This memo is intended for discussion only. Final production cost should be recalculated after confirming site conditions, utility tariff, facility scope, operating model, and offtake terms.